ENBC332: Homework #10

Due: Monday, 12/08/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 100

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:
e.g., calculating range.

```
# ---------------------------------------------------
Data <- trees$Height
my_range_func <- function(Data){
  Range <- max(Data) - min(Data)
  return(Range)
}
# ---------------------------------------------------
```

One-way ANOVA

Problem 1. A clinical trial compares four treatment groups. Each group has 10 patients ($n = 40$). The observed group means are:

Group 1: $\bar{x}_1 = 100$

Group 2: $\bar{x}_2 = 110$

Group 3: $\bar{x}_3 = 95$

Group 4: $\bar{x}_4 = 105$

Find the missing values in the ANOVA table below. (10 points)

Source	**SS**	**df**	**MS**	**F**
Between Groups				
Within Groups	1800			
Total				

Problem 2. You are studying the effect of different treatments on systolic blood pressure reduction (mmHg) after 8 weeks. Five groups are included: Control (no active treatment), Treatment_A, Treatment_B, Treatment_C, and Treatment_D. A dataset of 1000 patients has been collected. We are interested in **Reduction_mmHg** (negative means a drop in systolic BP). Use the give dataset medical_dataset.xlsx for the questions below. (90 points)

a) State the null and alternative hypotheses for a one-way ANOVA test. (5 points)

b) Compute the ANOVA table (sum of squares, degrees of freedom, mean squares), calculate the F-statistic, and find the critical F-critical at $\alpha = 0.05$. Show the step-by-step calculations of SST, SSB, SSW, df, MS, and F. (15 points)

c) Compute the p-value for the F-statistic and state your conclusion about whether there is a statistically significant difference among group means at $\alpha = 0.05$. (10 points)

d) If the ANOVA indicates significant differences, perform post-hoc test to determine which groups differ (name the test and briefly explain why). (20 points)

e) Write R code that computes the ANOVA (including the manual ANOVA table and F-critical) and prints the conclusion. (20 points)

f) Answer the following questions: (20 points)

1. Why is ANOVA preferred over running multiple independent t-tests when comparing more than two groups?
2. What does the F-statistic in ANOVA represent conceptually?
3. ANOVA tests whether group means differ, but it does not tell you which means differ. Why is this the case? What additional step is needed?
4. Why is the assumption of equal variances important in a one-way ANOVA? What can you do if this assumption is violated?
5. Explain the difference between statistical significance and practical (clinical) significance in the context of this medical blood pressure study.